Olufolajimi (Jimi) Oke

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Appointments

University of Massachusetts Amherst, MA

2025 – date *Associate Professor*, Civil and Environmental Engineering
2019 – 2025 *Assistant Professor*, Civil and Environmental Engineering
Director, Networks for Accessibility, Resilience & Sustainability Laboratory (NARS Lab)

Massachusetts Institute of Technology, Cambridge, MA

2016 – 2019 *Postdoctoral Associate*, Civil and Environmental Engineering (ITS Lab; PI: Moshe Ben-Akiva)
Project Manager, Future Urban Mobility, MIT Energy Initiative Mobility of the Future Study

The Pennington School, Pennington, NJ

2011 – 2012 *Faculty*: Department of Mathematics; Center for Learning
2010 – 2011 *Teaching Fellow*: Center for Learning

Education

Johns Hopkins University, Baltimore, MD

2016 *Doctor of Philosophy*: Civil Engineering (Advisor: Sauleh Siddiqui)
2014 *Master of Science in Engineering*: Civil Engineering

Williams College, Williamstown, MA

2010 *Bachelor of Arts*: Physics, Music
Honors Thesis in Physics (Advisor: Jefferson Strait)
Senior Recital, Classical Guitar (Teacher: Robert Phelps)

Fellowships

Massachusetts Institute of Technology

2019 Kaufman Teaching Certificate, *Teaching & Learning Lab*
2017 Postdoctoral Teaching Fellowship, *Department of Civil and Environmental Engineering*

Johns Hopkins University

2015 Gordon Croft Fellowship, *Environment, Energy, Sustainability & Health Institute (E^2SHI)*
2015 Teaching-as-Research Fellowship, *Center for the Integration of Research, Teaching and Learning*
2013 Educational Training Core Traineeship, *Global Obesity Prevention Center*
2012 Whiting School of Engineering Research Fellowship

Williams College

2008, 09 Summer Science Research Fellowship

External Grants

	Sponsor, Project Title (PI, Co-PIs)	*Role*	*Amount*
2026	**New England University Transportation Center** Investigating the relationship between transit characteristics and road safety outcomes *(J. Oke)*	PI	$25,000
2025	**New England University Transportation Center** How do mode-specific network metrics impact safety outcomes? *(J. Oke)*	PI	$75,000
2024 – 2025	**Connecticut South Central Regional Council of Governments** A predictive decision-making framework for robust and equitable climate action planning in the South Central Region of Connecticut *(J. Oke, C. Barchers)*	PI	$179,986
2024 – 2029	**National Science Foundation** Expeditions in Computing for Computational Decarbonization of Societal Infrastructures at Mesoscales *(P. Shenoy, A. Chien, M. Srivastava, Y. Agarwal, V. Sze)*	Co-I	$12,000,000
2024	**National Science Foundation** Conference: CET: Accelerating integrated Seabed Characterization for clean ENergy Transition (ASCENT) *(Z. Westgate, D. DeGroot, J. Brigham-Grette, J. Oke)*	Co-PI	$95,111
2024 – 2025	**New England University Transportation Center** Learning a spatial crash typology representation for analyzing and improving multimodal road safety in New England *(J. Oke)*	PI	$75,000
2023 – 2024	**Connecticut Metropolitan Council of Governments** Tracking regional emissions for climate action *(C. Barchers, J. Oke)*	Co-PI	$61,717
2023 – 2024	**Connecticut South Central Regional Council of Governments** Tracking regional emissions for climate action *(C. Barchers, J. Oke)*	Co-PI	$125,000
2023 – 2024	**Massachusetts Department of Transportation** Artificial intelligence framework for midblock crosswalk detection across Massachusetts *(J. Oke, Y. Xie, C. Ai, F. Tainter)*	PI	$99,998
2023 – 2026	**National Science Foundation** REU Site: Computing for an Equitable Energy Transition *(D. Irwin, J. Taneja)*	Co-I	$433,765
2023 – 2024	**Massachusetts Department of Transportation** Robust decision-making framework for sustainable operations and planning of MBTA rapid transit vehicles *(J. Oke, E. Christofa, E. Gonzales)*	PI	$125,000
2023 – 2026	**Environmental Protection Agency** Community-engaged Co-Design of Residential Electrification for a Just and Sustainable Energy Transition *(K. Harper, N. Caverly, C. Barchers, et al.)*	Co-I	$1,111,421
2021 – 2023	**Pioneer Valley Transit Authority** Pioneer Valley Transit Review and Improvement Planning Study (PV-TRIPS) *(C. Barchers, J. Oke)*	Co-PI	$137,719
2021 – 2022	**American Public Power Association** Cost-benefit analysis of co-deploying optical fiber and electric cabling underground *(J. Oke, A. Goldstein)*	PI	$191,197
2021 – 2023	**Pioneer Valley Transit Authority** Installation of On-Bus Mobile Ticket Validators and Development of an Innovative Origin-Destination-Transfer (ODX) Model *(J. Oke)*	PI	$103,952
2020	**National Science Foundation** America's Water Risk: Water System Data Pooling for climate vulnerability assessment and warning system *(U. Lall, C. Brown, S. Steinschneider, K. Kunkel)*	Co-I	$999,982
2020 – 2021	**Massachusetts Department of Transportation** Tracking and Reducing the Energy, Emissions and Costs of Rapid Transit Vehicles (TREEM) *(J. Oke, E. Christofa, E. Gonzales)*	PI	$122,522

Total External Grants	**$15,861,450**

Internal Grants

	Sponsor, Project Title (PI, Co-PIs)	*Role*	*Amount*
2025 – 2026	**Faculty Research Grant/Healey Endowment Grant** Novel public water system typology for effective deployment of innovative treatment technologies across the United States *(J. Oke, E. Kumpel)*	PI	$19,911
2024 – 2026	**Armstrong Fund for Science** Can schematic bicycle maps help reduce greenhouse gas emissions? *(J. Oke, E. Christofa)*	PI	$40,000
2021	**Brack Endowment Award for Research** Using Artificial Intelligence to Discover Defects and Predict Failure *(S. Gerasimidis, J. Oke)*	Co-PI	$15,032
	Total Internal Grants		**$74,943**

Awards

2024 2023 ASCE Outstanding Reviewer, *Journal of Infrastructure Systems*
2015 Article selected for Promotion, Elsevier, *Journal article "Tracking global bicycle ownership patterns"*
2015 Civil Engineering Graduate Service Award, *G.W.C. Whiting School of Engineering*
2013 Half-tuition Scholarship: "Modeling and Simulation of Transportation Networks", *MIT Professional Education Short Programs*
2010 Howard P. Stabler Prize in Physics, Williams College
2010 William W. Kleinhandler Prize for Excellence in Music, Williams College
2010 Sigma Xi Honors

Publications

Peer-Reviewed Journal Articles (42 published; 6 under review/revision; 5 in preparation)

[J53] Pranesh,* E. Kumpel, **J. Oke**, Intermittent water supply typology (**In preparation**) *Environmental Research Letters.*

[J52] P. Zhao,* E. Kumpel, **J. Oke**, Water violations typology (**In preparation**) *Environmental Science and Technology.*

[J51] C. Little,* **J. Oke**, E. Kumpel, Data-driven typology of US public water systems reveals infrastructure diversity beyond current regulatory classifications (**In preparation**) *Water Research.*

[J50] P. Zhao,* A. Weinman,** V. Rost-Nasshan,** M. Abdalazeem,* Z. Han,* M. Arabi,* and **J. Oke**, Road network typology of US metropolitan areas explains congestion, mobile emissions, and safety outcomes (**In preparation**).

[J49] C. S. Genis,* **J. Oke**, S. MacBeath, Drinking water treatment typology (**In preparation**) *Water Research.*

[J48] T. Nsubuga,* R. Vera, A. C. Marques, C. S. Lograsso, R. Cole, T. Hachey, B. DeJesus, J. Johnson, C. E. Hatch, T. Randhir, **J. Oke**, C. D. Guzman, Novel typology of monitoring stations reveals limited total phosphorus declines and hydropedological data gaps, (Under **revision**) *Environmental Research: Water.*

[J47] M. Abdalazeem,* C. Barchers, E. Christofa, **J. Oke**, Towards unified design rules for schematic micromobility maps (Under **review**) *Environment and Planning B: Urban Analytics and City Science.*

[J46] H. Bell,** N. Bohlke,** M. Abdalazeem,* **J. Oke**, E. Christofa, Three-decade systematic review highlights enduring barriers to cycling and reveals rural–urban evidence gap (Under **review**) *Journal Transport and Health.*

[J45] P. Zhao,* M. Arabi,* D. Rajagopal, **J. Oke**, A Review of Agent-Based Modeling, AI, and Emissions Frameworks for Freight Decarbonization (Under **review**) *Next Sustainability.*

[J44] P. Zhao,* **J. Oke**, Urban typologies emerge as a viable framework for informing mobility, sustainability and urban planning (Under **review**) *Environmental Research Letters.*

[J43] M. Azhdari,* **J. Oke**, E. Gonzales, Unsupervised Discovery of Electric Bus Driving Maneuver regimes from Vehicle Trajectory Data Using Deep Autoencoder and Gaussian Mixture Model, (Under **2nd review**) *Data Science for Transportation.*

[J42] M. Abdalazeem,* A. Ruger,** E. Christofa, **J. Oke** (2026). Can metro-style schematic maps induce micromobility ridership? A Bayesian latent class ordered logistic analysis (**Accepted**) *Data Science for Transportation.*

[J41] Y. Xie, X. Yan, Z. Bhuyan, **J. Oke**, G. Wu (2026). A two-stage detection and segmentation framework for pedestrian crosswalk inventory and condition assessment from aerial imagery, *Engineering Applications of Artificial Intelligence*, 182 (1), 115851. DOI

[J40] E. Kostopoulou,* T. Oke, **J. Oke**, E. Christofa (2026). A Scenario Discovery Approach to Transit Network Optimization for Improved Access in Areas of Persistent Poverty, *Data Science for Transportation*, 8 (2), 26. DOI

[J39] P. Zhao,* **J. Oke** (2026). Early action in transportation, electricity grid, commercial and residential buildings is essential for deep and near net-zero decarbonization, *Next Sustainability*, 8 (100375). DOI

[J38] N. Kumar,* Y. Zhang, N. Wiedemann, **J. Oke**, M. Raubal (2026). A multiscale interpretability framework for identifying actionable road network features to mitigate congestion in highly congested cities, *Scientific Reports.* DOI

[J37] A. Ruger,** M. Abdalazeem,* E. Christofa, **J. Oke** (2026). A Multi-Filter Approach to Bot Detection in Incentivized Online Surveys, *Data Science for Transportation*, 8 (2) 21. DOI

[J36] M. Azhdari,* **J. Oke**, E. Gonzales (2026). A Functional Bus Stop Typology: Explainable Machine Learning on Transit Data to Inform Policymaking, *Data Science for Transportation*, 8 (2), 13. DOI

[J35] P. Zhao,* J. Mega, M. Abdalazeem,* M. Arabi,* C. Barchers, **J. Oke** (2026). Scenario discovery yields robust multi-level pathways for regional decarbonization planning, *Environmental Research Communications*, 8 (3), 031008. DOI

[J34] M. Abdalazeem,* G. Altayb,** E. Christofa, **J. Oke**§ (2026). Maps are good but are simpler maps better? Insights on urban bicycling in the US *Environmental Research Communications*, 8 (2), 025004. DOI

[J33] M. Arabi,* **J. Oke** (2026). Typology of U.S. transit systems reveals energy data gap and potential for diesel-based emissions reductions, *Environmental Research Letters*, 21, 024014. DOI

[J32] Z. Han,* E. Gonzales, E. Christofa, **J. Oke** (2026). Robust forecasting framework for energy-informed planning in urban rail transit systems, *Data Science for Transportation*, 8(2). DOI

* Graduate advisee/mentee **Undergraduate advisee/mentee **Undergraduate advisee/mentee * Graduate advisee/mentee **Undergraduate advisee/mentee § Corresponding author

[J31] T. Oke,* A. Pate, F. Tainter, **J. Oke**, M. Knodler (2025). Bus stop typology reveals crash risk environments, *Data Science for Transportation*, 7(3). DOI

[J30] Z. Han,* **J. Oke** (2025). Novel typology of fatal crash behaviors via natural language processing, *Data Science for Transportation*, 7(22). DOI

[J29] M. Arabi,* T. Oke, E. D. Baker, **J. Oke** (2025). Harnessing hybrid buses in the near term leads to faster transit decarbonization, *iScience*, 28(10), 113567. DOI

[J28] P. Zhao,* **J. Oke** (2025). US metro typology illuminates per-capita mobile greenhouse gas emissions impacts and informs mitigation strategies, *Environmental Research Letters*, 20, 084075. DOI

[J27] M. Abdalazeem,* **J. Oke** (2025). Roadway Crash Typology of Census Tracts Enables Targeted Interventions via Interpretable Machine Learning, *Data Science for Transportation* 7(14). DOI

[J26] D. Irwin, P. Shenoy, M. Hajiesmaili, W. A. Hanafy, **J. Oke**, R. Sitaraman, Y. Agarwal, G. Gordon, Z. Kolter, D. Rajagopal, M. Srivastava, V. Sze, P. Donti, A. Chien, J. Birge, A. Hortacsu, L. Roald (2025). A Vision for Computational Decarbonization of Societal Infrastructure, *IEEE Internet Computing* 29(2), 27–35. DOI

[J25] M. Arabi,* A. Apostolov,* A. Pattabhiraman, A. Goldstein, M. Bloomberg, J. Taneja, E. D. Baker, **J. Oke** (2025). Benefits of rapidly co-undergrounding electric and broadband lines outweigh costs, *Cell Reports Sustainability*, **2**(3), 100334. DOI

[J24] M. Abdalazeem,* **J. Oke** (2024). Enhanced seasonal typology-informed transit trip chaining via mobile boarding and survey data, *Data Science for Transportation*, **6**(3), 1-24. DOI

[J23] M. Arabi,* S. Gerasimidis, C. Barchers, **J. Oke** (2024). Resilience in multilayer transportation infrastructure networks: a review and conceptual framework for equity-based assessment, *Sustainable and Resilient Infrastructure*, 1-24. DOI

[J22] D. Sarigiannis, M. Atzemi, **J. Oke**, E. Christofa, S. Gerasimidis (2024). Feature Engineering and Decision Trees for Predicting High Crash-Risk Locations using Roadway Indicators, *Transportation Research Record*, DOI

[J21] M. Arabi,* **J. Oke** (2024). Comprehensive energy modeling framework for multi-powertrain bus transit systems, *Transportation Research Record* **2678**(1): 707-720 DOI

[J20] Z. Han,* E. Gonzales, E. Christofa, **J. Oke** (2024). Line-specific energy modeling framework for urban rail transit systems: A case study of Boston, *Transportation Research Record* **2678**(1): 150-164 DOI

[J19] M. Abdalazeem,* **J. Oke** (2023). Extracting spatiotemporal bus passenger trip typologies from noisy mobile ticketing boarding data, *Data Science for Transportation* **5**(3), 20. DOI

[J18] A. Apostolov,* **J. Oke**, R. Suttle, S. Arwade, B. Kane (2023). Predicting tree failure likelihood for utility risk mitigation via a convolutional neural network, *Sustainable and Resilient Infrastructure*, **8**(6), 572-588 DOI

[J17] H. Chung,** E. Kumpel, **J. Oke**§ (2023). Drinking water accessibility typologies in low- and middle-income countries, *Environmental Research Letters* **18** 025009. DOI

[J16] S. Goodarzi,* H. F. Kashani, A. Saeedi, **J. Oke**, C. L. Ho (2023). Stochastic analysis for estimating track geometry degradation rates based on GPR and LiDAR data, *Construction and Building Materials*, 369: 130591. DOI

[J15] M. Mohammed,* **J. Oke** (2023). Origin-destination inference in public transportation systems: a comprehensive review, *International Journal of Transportation Science and Technology*, 12 (1), 315-328.

* Graduate advisee/mentee * Graduate advisee/mentee **Undergraduate advisee/mentee * Graduate advisee/mentee

DOI

[J14] Z. Han,* E. Gonzales, E. Christofa, **J. Oke** (2022). Modeling system-wide urban rail transit energy consumption: A case study of Boston, *Transportation Research Record*, 2676(12): 627-640. DOI

[J13] S. Goodarzi,* H. F. Kashani, **J. Oke**, C. L. Ho (2022). Data-driven methods to predict track degradation: a case study, *Construction and Building Materials*, 344: 128166. DOI

[J12] N. Kumar, **J. Oke**, B. Nahmias-Biran (2021). Activity-based epidemic propagation and contact network scaling in auto-dependent metropolitan areas, *Scientific Reports*, 11(1): 1-14. DOI

[J11] B. Nahmias-Biran, **J. B. Oke**, N. Kumar (2021). Who Benefits from AVs? Social Implications of Autonomous Vehicle Policies in Full-Scale Cities, *Transportation Research Part A*, 154: 92-107. DOI

[J10] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, C. L. Azevedo, M. Ben-Akiva (2021). Evaluating the impacts of shared automated mobility on-demand: an activity-based accessibility approach, *Transportation*, 48: 1613–1638. DOI

[J9] **J. B. Oke**, A. P. Akkinepally, S. Chen, Y. Xie, Y. M. Aboutaleb,* C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva (2020). Evaluating systemic effects of automated on-demand services through large-scale, agent-based simulation of auto-dependent, prototype cities, *Transportation Research Part A*, 140: 98-126. DOI

[J8] D. A. Martinez, J. Cai, **J. B. Oke**, A. Jarrell, F. Feijoo, J. Appelbaum, E. Klein, S. Barnes, S. R. Levin (2020). Where is my Infusion Pump? Harnessing Network Dynamics for Improved Hospital Equipment Fleet Management, *Journal of the American Medical Informatics Association*, 27(6): 884-892. DOI

[J7] **J. B. Oke**, Y. M. Aboutaleb,* C. L. Azevedo, Y. Han, A. Akkinepally, P. C. Zegras, J. Ferreira, M. E. Ben-Akiva (2019). A novel global urban typology framework for sustainable mobility futures, *Environmental Research Letters*, 14(9): 95006. DOI

[J6] B. Nahmias-Biran, **J. B. Oke**, C. L. Azevedo, N. Kumar, A. Araldo, K. Basak, R. Seshadri, M. Ben-Akiva (2019). From traditional to automated mobility on demand: a comprehensive framework for modeling mobility on demand services in SimMobility, *Transportation Research Record*, 2673(12): 15-29. DOI

[J5] **O. Oke**, D. Huppmann, M. Marshall,** R. Poulton,** S. Siddiqui (2019). Multimodal transportation flows in energy networks with an application to crude oil markets, *Networks and Spatial Economics*, 19(2): 521-555. DOI

[J4] **O. Oke**, K. Bhalla, D. C. Love, S. Siddiqui (2018). Spatial associations in global bicycle ownership, *Annals of Operations Research*, 263: 529-549. DOI

[J3] **O. Oke**, K. Bhalla, D. C. Love, S. Siddiqui (2015). Tracking global bicycle ownership patterns, *Journal of Transport and Health*, 2(4): 490-501. DOI

[J2] **O. Oke**, S. Siddiqui (2015). Efficient automated schematic map drawing using multiobjective mixed integer programming, *Computers and Operations Research*, 61:1-17. DOI

[J1] C. Chudzicki, **O. Oke**, W. K. Wootters (2010). Entanglement and Composite Bosons, *Physical Review Letters*, 104(7):070402. DOI

Reports, Non-Peer Reviewed Articles and Theses

[R10] Z. Westgate, D. DeGroot, J. Brigham-Grette, J. Oke, *Navigating the ASCENT: Outcomes From an NSF Workshop on Offshore Wind Seabed Characterization*, GEOSTRATA, Oct/Nov 2025.

[R9] J. Oke, *Typology-Informed Solutions for Sustainable and Accessible Transportation Networks*, in: Transportation Research Circular E-C301, "Access Management: Safety and Design from the Curb to the Interchange for All Users," May 2025.

[R8] Z. Han, E. Christofa, E. Gonzales, **J. Oke**, *Robust Decision-Making Framework for Sustainable Opera-*

tions and Planning of MBTA Rapid Transit Vehicles, 24-063, Massachusetts. Dept. of Transportation. Office of Transportation Planning, September 2024.

[R7] A. Apostolov, Y. Xie, C. Ai, F. Tainter, **J. Oke**, *Artificial Intelligence Framework for Crosswalk Detection Across Massachusetts*, 20-024, Massachusetts. Dept. of Transportation. Office of Transportation Planning, February 2024.

[R6] **J. Oke**, E. Christofa, E. Gonzales, Z. Han, S. Donaghy, *Tracking the Energy and Emissions of MBTA Rapid Transit Vehicles*, 21-023, Massachusetts. Dept. of Transportation. Office of Transportation Planning, August 2021.

[R5] W. H. Green et al., *Insights Into Future Mobility: A Report from the Mobility of the Future Study*, MIT Energy Initiative, Cambridge, MA, November 2019.

[R4] **O. Oke**, *Network modeling and optimization for energy and sustainable transit*, Doctoral Dissertation in Civil Engineering, Johns Hopkins University, May 2016.

[R3] **J. Oke**, S. Siddiqui, K. Bhalla, D.C. Love, J. De Vito, M. Van Doren, Max Marshall, *Making Baltimore More Bike Friendly*, Department of Civil Engineering, Johns Hopkins University, March 2014.

[R2] **O. Oke**, *Bicycling in Baltimore: key concerns*, Report (submitted to Baltimore DOT), Department of Civil Engineering, Johns Hopkins University, January 2013.

[R1] **O. Oke**, *A nonlinear optical loop mirror modelocked fiber laser*, Honors Thesis in Physics, Williams College, May 2010.

Conference Papers & Extended Abstracts [Peer Reviewed]

[C18] M. Abdalazeem,* T. Oke, **J. Oke**, *A Typology-Informed Origin-Destination-Transfer Model for a Bus Transit Network Using Mobile Ticketing Data*, TRB Annual Meeting (2025).

[C17] M. Abdalazeem,* **J. Oke**, *A Spatial Typology Analysis of Crash Characteristics Across 2,480 Census Tracts*, TRB Annual Meeting (2025).

[C16] Z. Han,* E. Gonzales, E. Christofa, **J. Oke**, *Energy forecasting framework for austainable operations planning in urban rail transit systems*, TRB Annual Meeting (2025).

[C15] P. Zhao,* **J. Oke**, *A Novel Mobility Typology of 355 U.S. Metropolitan Areas and its Impacts on Greenhouse Gas Emissions.*, TRB Annual Meeting (2025).

[C14] M. Arabi,* **T. Oke**, **J. Oke**, *Sustainable Transition to Zero-Emissions Fleet Requires Both Electric and Hybrid Buses in the Near-Term*, TRB Annual Meeting (2025).

[C13] P. Zhao,* **J. Oke**, *Tracking and Predicting Regional Greenhouse Gas Emissions: A Case Study of Connecticut.*, AIChE Annual Meeting (2024).

[C12] Z. Han,* E. Gonzales, E. Christofa, **J. Oke**, *Energy forecasting and scenario planning for sustainable decision-making in urban rail transit systems*, TRB Annual Meeting (2024).

[C11] Z. Han,* E. Gonzales, E. Christofa, **J. Oke**, *Line-specific energy modeling framework for urban rail transit systems: A case study of Boston*, TRB Annual Meeting (2023).

[C10] M. Arabi,* **J. Oke**, *System-wide Energy Modeling of Bus Transit Systems: A Case Study of The Pioneer Valley*, TRB Annual Meeting (2023).

[C9] M. Mohammed,* **J. Oke**, *Spatiotemporal Trip Chaining Framework for Open Mobile Fare Collection Systems*, TRB Annual Meeting (2023).

[C8] N. Apostolov,* **J. Oke**, *Retrospective Investigation of Country Activity and Mobility Patterns and Their Interdependencies on Early-onset COVID-19 Outcomes*, TRB Annual Meeting (2023).

[C7] S. Goodarzi,* H. F. Kashani, **J. Oke**, C. L. Ho, *Stochastic Analysis for Estimating Track Geometry Degradation Rates Based on GPR and LiDAR Data*, TRB Annual Meeting (2023).

[C6] Z. Han,* E. Gonzales, E. Christofa, **J. Oke**, *Modeling system-wide urban rail transit energy consumption: A case study of Boston*, TRB Annual Meeting (2022).

[C5] M. Mohammed,* **J. Oke**, *Trip pattern typologies in the Pioneer Valley bus transit system*, TRB Annual Meeting (2022).

[C4] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, *Activity-based contact network scaling and epidemic propagation in metropolitan areas*, TRB Annual Meeting (2021).

[C3] B. Nahmias-Biran, **J. B. Oke**, N. Kumar, A. P. Akkinepally, C. L. Azevedo, P. C. Zegras, J. Ferreira, M. Ben-Akiva, *Who Benefits from AVs? Equity Aspects of Autonomous Vehicles Policies in a Full-Scale Prototype Cities*, TRB Annual Meeting (2020).

[C2] E. Gross,* **J. Oke**, A. P. Akkinepally, B. Nahmias-Biran, C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva, *Accessibility and energy consumption evaluation under different strategies of mobility on-demand deployment*, TRB Annual Meeting (2019).

[C1] Y. Han, **J. Oke**, S. Hua,* J. Zhou,* C. L. Azevedo, C. Zegras, J. Ferreira, M. Ben-Akiva, *Global urban typology discovery with a latent class choice model*, TRB Annual Meeting (2018).

COMMUNICATIONS

Invited Presentations

[IP18] *Typology-informed solutions for sustainable and accessible infrastructure networks*, 4th International Conference on Access Management, Boston, MA, June 2024.

[IP17] *Machine learning-enhanced decision-making frameworks for sustainable mobility systems*, 2024 Emerging Technologies in Transportation Seminar, University of Florida Gainesville, FL, March 2024.

[IP16] *Mobility and epidemics in urban typologies*, Diversity, Equity and Inclusion Seminar, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, MA, June 2020.

[IP15] *Analysis of future mobility on-demand systems in global urban typologies*, INFORMS Annual Meeting, Seattle, WA, October 2019.

[IP14] *New urban typologies for sustainable mobility*, Department of Civil and Environmental Engineering, University of Illinois Urbana-Champaign, IL, February 2019.

[IP13] *New urban typologies for sustainable mobility*, Department of Civil and Environmental Engineering, University of Massachusetts Amherst, MA, February 2019.

[IP12] *Discovering new urban typologies*, Department of Civil and Environmental Engineering, University of Pittsburgh, PA, January 2019.

[IP11] *Discovering sustainable future urban mobility policies via simulation of prototype cities from global urban typologies*, Systematizing and upscaling urban solutions for climate change mitigation, Berlin, Germany, September 2018.

[IP10] *Discovering Robust Urban Mobility Futures via Agent Based Simulation in Prototype Cities*, 7th TRB Innovations in Travel Modeling Conference, Atlanta, GA, June 2018.

[IP9] *Need for and Uses of Risk Analysis: Technical approaches from the university perspective*, Risk Analysis Workshop, 7th TRB Innovations in Travel Modeling Conference, Atlanta, GA, June 2018.

[IP8] *Urban mobility simulation for scenario discovery in globally representative prototypes*, Physics Colloquium, Department of Physics, Williams College, MA, October 2017.

[IP7] *Exploring sustainable mobility strategies in future cities*, Log Lunch Series, Center for Environmental Studies, Williams College, MA, October 2017.

[IP6] *A Crude Oil Market Model for the United States*, INFORMS Annual Meeting, Philadelphia, PA, Novem-

ber 2015.

[IP5] *An Equilibrium Model of the US Crude Oil Market*, the 22nd International Symposium on Mathematical Programming, Pittsburgh, PA, July 2015.

[IP4] *Redefining Infrastructural Space*, Environment, Energy, Sustainability & Health Institute Art/Science Roundtable, Maryland Institute College of Art, Baltimore, May 2015.

[IP3] *Tracking Global Bicycle Ownership Patterns*, INFORMS Data Mining and Analytics Workshop, San Francisco, CA, November 2014.

[IP2] *An efficient automated multiobjective programming approach to map schematization*, Systems/Policy/Energy Seminar, Johns Hopkins University, March 2014.

[IP1] *A Mixed-integer Programming Tool for Creating Effective Schematic Urban Transit Maps*, INFORMS Annual Meeting, Minneapolis, MN, October 2013.

Contributed Presentations

[CP9] *Metropolitan Area Road Network Typologies and Their Implications for Sustainable Mobility*, INFORMS Annual Meeting, Phoenix, AZ, October 2023.

[CP8] *Assessing the energy impacts of automated on-demand service deployment strategies in sprawling, auto-dependent cities*, Transatlantic Infraday Conference, Federal Energy Research Commission, Washington, DC, November 2018.

[CP7] *Discovering Urban Typologies For Future Mobility Scenarios In Prototype Cities*, INFORMS Annual Meeting, Houston, TX, October 2017.

[CP6] *Analyzing United States Crude Oil Flows*, Transatlantic Infraday Conference, Federal Energy Research Commission, Washington, DC, October 2015.

[CP5] *An Oil Market Model for the United States*, Modeling and Optimization: Theory and Applications, Lehigh University Department of Industrial and Systems Engineering, Bethlehem, PA, July 2015.

[CP4] *Solving the Crude on Rail Problem using an Equilibrium Model of the US Crude Oil Market*, Critical Infrastructure Symposium, Linthicum, MD, April 2015.

[CP3] *Global Bicycle Availability*, INFORMS Annual Meeting, San Francisco, CA, November 2014.

[CP2] *Multiobjective optimization for automatic schematic map drawing*, Modeling and Optimization: Theory and Applications, Lehigh University Department of Industrial and Systems Engineering, Bethlehem, PA, August 2014.

[CP1] *Schematic map automation and optimization*, Civil Engineering Graduate Seminar, Johns Hopkins University, November 2013.

Teaching

University of Massachusetts Amherst, MA

Instructor, Department of Civil and Environmental Engineering

Spring 2026 Advanced Probabilistic Machine Learning (Graduate)
5 students

Fall 2025 Probabilistic Machine Learning (Graduate)
13 students

Fall 2025 Probability & Statistics in Civil Engineering (Undergraduate)
161 students

Spring 2025 Machine Learning Foundations and Applications (Graduate/Undergraduate)

18 students

Fall 2024 Probability & Statistics in Civil Engineering (Undergraduate)
126 students

Spring 2024 Advanced Probabilistic Machine Learning (Graduate)
Newly developed course; 9 students

Fall 2023 Machine Learning Foundations and Applications (Graduate/Undergraduate)
Newly developed course; 12 students

Spring 2023 Data Mining and Machine Learning for Engineers (Graduate/Undergraduate)
18 students

Spring 2022 Data Mining and Machine Learning for Engineers (Graduate/Undergraduate)
15 students

Fall 2021 Probability & Statistics in Civil Engineering (Undergraduate)
116 students

Spring 2021 Data Mining and Machine Learning for Engineers (Graduate/Undergraduate)
9 students

Fall 2020 Probability & Statistics in Civil Engineering (Undergraduate)
60 students

Spring 2020 Data Mining and Machine Learning for Engineers (Graduate/Undergraduate)
16 students; created new course

Fall 2019 Probability & Statistics in Civil Engineering (Undergraduate)
127 students; introduced MATLAB programming assignments

Massachusetts Institute of Technology, MA

Teaching Assistant, Department of Civil and Environmental Engineering

Fall 2018 Transportation Systems Analysis: Demand and Economics (Graduate)
15 students, recitations, content development

Summer 2018 Modeling and Simulation of Transportation Networks (Professional)
content development

Fall 2017 Transportation Systems Analysis: Demand and Economics (Graduate)
25 students, recitations, content development, assessments

Teaching Fellow, Department of Civil and Environmental Engineering

Spring 2016 Multivariate Data Analysis (Undergraduate)
16 students, recitations, lectures, Jupyter notebook development, innovative assessments, project

Johns Hopkins University, MD

Instructor, Hopkins Engineering Applications & Research Tutorials Program

Fall 2015 Reality Distortion: The Impact and Automation of Schematic Maps (Undergraduate)
3 students, content development: GAMS, Python, optimization, visualization

Teaching-as-Research Fellow, Department of Civil Engineering

Spring 2015 Probability and Statistics in Civil Engineering (Undergraduate)
97 students, designed 12 applied MATLAB projects, surveys & focus group to measure impact

Teaching Assistant, Department of Civil Engineering

Oct 2013 Optimization and Equilibrium Modeling in Systems Engineering (Graduate, day-long)
11 students, GAMS installation and programming help

Spring 2013 Probability and Statistics in Civil Engineering (Undergraduate)
55 students, office hours, wrote and graded quizzes & exam problems
Fall 2012 Statics and Mechanics of Materials (Undergraduate)
110 students, taught 10 of 12 lab sections, assessed lab reports, course grade manager

The Pennington School, NJ

2011 – 2012 **Faculty Member**, Mathematics Department
Algebra II Honors, Precalculus (Course Leader)
developed innovative assessments, redesigned Precalculus curriculum

2010 – 2012 **Teaching Fellow – Faculty Member**, Center for Learning
Algebra II, Math Skills Tutorial, Writing Skills Tutorial, Communication Skills
small classes, students with learning differences, focused mentorship

Williams College, MA

Physics Tutor, Office of Academic Resources
2008 – 2010 Peer Tutor Program
one-on-one appointments several times a week
2007 – 2010 Math and Science Resource Center
walk-in/group sessions, 2-hour shifts, 2 – 4 times a week

Teaching Assistant, Department of Physics
Spring 2009 "Waves and Optics"
grading, homework assistance
Fall 2008 "Particles and Waves, Enriched"
grading, homework assistance

Teaching Assistant, Department of Art
Fall 2007 Drawing
managed studio, curated student work

Service

Graduate Research Mentorship

2023–date Peiyao Zhao (PhD, UMass Amherst)
2021–date Mohammed Abdalazeem Mohammed (MSCE*/PhD, UMass Amherst)
2021–2025 Mahsa Arabi (PhD, UMass Amherst)
2022–2025 Nishant Kumar (PhD, ETH Zurich; secondary advisor)
2020–2025 Zhuo Han (MSCE*/PhD, UMass Amherst)
2020–2024 Atanas Apostolov (MSCE*, UMass Amherst)
2018–19 Yifei Xie (MST†, MIT), Siyu Chen (MST†, MIT)
2017–18 Eytan Gross (MST†, MIT), Iveel Tsogsuren (MEng, MIT)
2017–19 Youssef Medhat (MST†, MIT)

* Masters of Science in Civil Engineering
† Masters of Science in Transportation

Summer 2017 Jin (Jasmine) Zhou (MA, Columbia University)
Spring 2017 Scott Foster (Leaders for Global Operations Fellow, MIT)
Fall 2016 Akshay Padmanabha (MEng, MIT)
2016–17 Sean Hua (MEng, MIT); Michael Choi (MEng, MIT)

Undergraduate Research Mentorship

Summer 2026 Julianna Demers (BS, Industrial and Management Engineering, Rensselaer Polytechnic Institute)
Spring 2026 Henry Bell (BS, Civil Engineering, UMass Amherst)
Summer 2025 Andrew Ruger (BA, Economics, Grinnell College)
Spring 2025 Nicholas Bohlke (BS, Civil Engineering, UMass Amherst)
Summer 2024 Geehan Altayb (BS, Computer Science, Howard University)
Summer 2023 Vivian Rost-Nasshan (BS, Industrial and Management Engineering, Rensselaer Polytechnic Institute)
2022-2023 Alexa Weinman (BSE 2023, Civil and Environmental Engineering, UMass Amherst)
2020-2021 Hichul Chung (BSE 2021, Mechanical Engineering, UMass Amherst)
Summer 2020 Adegbola Fayemi (College of Wooster)
2017–18 Sharlene Chiu (Super UROP‡, MIT)
Spring 2017 Joseph Noszek (UROP‡, MIT), Abenezer Samuel (UROP‡, MIT)
Winter 2017 Gabriel Madonna (Mini UROP‡, MIT)
2013–16 Max Marshall (JHU)
2014–15 Ricky Poulton (JHU)
2013–14 Molly Van Doren (JHU)

Dissertation Committees (candidates at UMass except where noted)

2025–date Khotso Selialia (PhD candidate, Electrical and Computer Engineering)
2025–2026 Efthymia Kostopoulou (PhD candidate, Civil Engineering)
2024–2026 Timothy Nsubuga (PhD candidate, Environmental Engineering)
2024–2025 Guocheng Jiang (PhD candidate, Civil Engineering)
2022–2025 Nishant Kumar (PhD, ETH Zurich)
2022–2023 Saeed Goodarzi (PhD, Civil Engineering, 2023)
2021–22 Brent Scott (PhD, Kinesiology, 2022)
2021 Peetak Mitra (PhD, Mechanical Engineering, 2021)
2019–20 Haralampos Sipetas (PhD, Civil Engineering, 2020)

Internal Service, Department and School/College

University of Massachusetts Amherst
2025–2026 Search Committee Member, Department of Civil and Environmental Engineering
Fall 2025 Transportation Engineering Program Seminar Coordinator
Fall 2023 Transportation Engineering Program Seminar Coordinator
2023–2024 Search Committee Member, Department of Mechanical and Industrial Engineering

Civil & Environmental Engineering Postdoc Committee, Massachusetts Institute of Technology
2016–17 Member

‡Undergraduate Research Opportunities Program

Johns Hopkins University

2015–16 Graduate Representative, Homewood Gradaute Board, Whiting School of Engineering
2015–16 President, Civil Engineering Graduate Association
2014–15 Social Coordinator, Civil Engineering Graduate Association
2014 Founded the Civil Engineering Graduate Association
2014–15 Graduate Seminar (Systems) Cochair, Department of Civil Engineering
2013–14 Graduate Seminar Committee Member, Department of Civil Engineering

Editorial Board

2025–date Editorial Board Member, Journal of Infrastructure Systems

Manuscript Review

2026 Transportation Research Record, Scientific Reports, Discover Cities, Nature Communications, Data Science for Transportation, Environmental Research Letters, Journal of Infrastructure Systems
2025 Reliability Engineering & System Safety, Discover Cities, Data Science for Transportation, Journal of Infrastructure Systems; Transportation Research Record; Energy Research & Social Science; Engineering Research Express; Transportation; Scientific Reports
2024 Transportmetrica A: Transport Science; Applied Network Science; Transportation Planning and Technology; Transportation; Journal of Urban Planning and Development; Urban Forestry & Urban Greening; Measurement Science & Technology; Transportation Research Board
2023 Transportation Research: Part A, Part D; Journal of Infrastructure Systems; Transportation Research Board; Simulation Modelling Practice and Theory; Data Science for Transportation; Open Research Europe; Cogent Social Sciences; Scientia Iranica
2022 Transportation Research: Part A, Part C, Part D; Transportation Research Board; Energy Journal; Journal of the American Medical Informatics Association; Energy Research & Social Science
2021 Energy Research & Social Science; Transportation Research: Part C; Journal of Urban Planning and Development
2020 Transportation Research Record
2019 Transportation Research Board
2018 Transportation Research: Part A; International Journal of Geographical Information Science; Journal of Transport & Health
2017 International Journal of Sustainable Transportation; Computational Optimization and Applications
2016 Optimization and Engineering
2015 IEEE Transactions on Power Systems
2014 INFORMS Data Mining & Analytics Workshop

Conference Organization

2023 Session Chair, INFORMS Annual Meeting
2017 Session Chair, INFORMS Annual Meeting

Proposal Review

Jun 2026 Panelist, Environmental Sustainability Review Panel, National Science Foundation
Mar 2026 Panelist, Environmental Sustainability Review Panel, National Science Foundation
Sep 2025 Panelist, Environmental Sustainability Review Panel, National Science Foundation

Jul 2024	Peer Reviewer, New England University Transportation Center Year 2 Proposals
Dec 2024	Panelist, Civil Infrastructure Systems (CIS) Engineering Research Initiation (ERI) Review Panel, National Science Foundation
Dec 2023	Fastlane Reviewer, Environmental Sustainability Program, National Science Foundation
Oct 2023	Peer Reviewer, New England University Transportation Center Year 1 Proposals
Nov 2022	Panelist, LEAP HI Infrastructural Systems Preliminary Panel, National Science Foundation

Other Outreach

Summer 2025	Seminar Speaker, UMass Turing Summer Program
Fall 2023	Reviewer, Council of University Transportation Centers Pikarsky Masters Student Award
2014 – 16	Graduate Student Mentor, STEM Achievement in Baltimore Elementary Schools (SABES)
2015	Contributor, Baltimore City Bicycle Master Plan 2015
2013	Project co-coordinator, "Making Baltimore More Bicycle Friendly"

Academic Affiliations

American Society of Civil Engineers (ASCE)
Institute for Operations Research and Management Sciences (INFORMS)
Transportation Research Board
- Member, Transportation Energy Committee (2023–2025)
- Member, Critical Infrastructure Protection Committee (2023–2025)

Transport and Health Science Group—North America (since 2025)